

SMICES | 2026–2027

AI for Cybersecurity

C1 | Introduction & Ethics

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7 October 2026



National University of Science and Technology POLITEHNICA Bucharest
Faculty of Electronics, Telecommunications and Information Technology

Course organisation



Lectures

Ana Nicolae

in person

Deep learning foundations and AI principles

Practical sessions

Vlad Vasilescu

Online

Applied work with cybersecurity datasets

Teaching sessions are on Wednesdays @ 17:00. Lecture and laboratory slots alternate as shown next.

All materials are available [here](#).

Assessment: 100 points



Points	Component	What you do
40	Lab homework	Choose one of the three homeworks ; present and discuss it in week 13 or 14 .
30	Midterm exam	Demonstrate understanding of the first part of the course.
30	Final exam	Demonstrate understanding of the final part of the course.

AI generated homework without understanding will be scored with 0 points.

Exams and presentation dates



Assessment	When	Scope / arrangement
Written midterm (30 points)	25 November 2026 Week 9	Linear networks; Optimisation; DNNs; simple CNNs.
Lab homework (40 points)	13 or 20 January 2027 Weeks 13 / 14	Presentation of one selected homework; individual slot allocation to follow.
Final exam (30 points)	Exact date TBA Exam window: 23 Jan–12 Feb 2027	CNNs and RNNs, as specified in the supplied schedule.

All written evaluation will be closed book, no AI permitted.

Wednesday schedule: September–November



Wk	Date	Type	Session
1	30 Sep	–	No class
2	7 Oct	C1	Introduction and Ethics of AI
3	14 Oct	C2	Linear networks and optimisation
4	21 Oct	–	No class
5	28 Oct	C3	Deep neural networks
6	4 Nov	L1	DNNs: optimisation and explainability
7	11 Nov	C4	Convolutional neural networks
8	18 Nov	L2	CNNs, transfer learning and a simple attack
9	25 Nov	Exam	Written midterm (30 points)

C = lecture; L = practical session.

Wednesday schedule: December–January



Wk	Date	Type	Session
10	2 Dec	L3	CNN applications and attacks
11	9 Dec	C6	Recurrent neural networks
12	16 Dec	L4	RNNs and malware API-call sequences
–	23, 30 Dec; 6 Jan	Break	Winter break: 19 Dec–10 Jan
13	13 Jan	Lab test	Selected-homework presentations
14	20 Jan	Lab test	Selected-homework presentations
–	TBA	Exam	Final exam (30 points)

Deep learning foundations

- Ethics of AI
- Linear networks and optimisation
- Fully connected neural networks
- Convolutional neural networks
- Recurrent neural networks

Today: introduction and ethics

1. What is AI, and why does it matter?
2. Algorithmic bias
3. A broader ethical perspective
4. High-risk applications
5. Building ethical and safe AI

From foundational models to responsible cybersecurity applications.

Signal Modelling and Analysis Lab

- **Robust neural networks:** resistance to adversarial perturbations, noise and distribution shifts; reliability and interpretability.
- **Audio analysis:** source separation, denoising, spatial audio and up-mixing.
- **Biomedical analysis:** learning from EEG and other physiological signals.

Connection to this course

Robustness and interpretable decisions are research questions and practical requirements for trustworthy security systems.

CAMPUS Research Institute, POLITEHNICA Bucharest; academic affiliation with ETTI.

www.sigmalab.ro

Why Artificial Intelligence?



“AI is the new electricity.”

Andrew Ng

- Electricity transformed transportation, manufacturing, healthcare and communications.
- AI can also transform a wide range of activities.
- This change comes with substantial challenges.

In cybersecurity, should faster decisions always mean more autonomous decisions?

Key features of AI



Artificial intelligence

The study and engineering of computer systems that perform tasks associated with intelligent behaviour: perception, reasoning, learning and decision-making.

Autonomy Performing tasks without constant guidance by a user.

Adaptability Improving performance through experience or learning.

Machine learning Learning patterns from data rather than explicitly programming every decision rule.

Autonomy and adaptation are degrees of capability, not guarantees of intelligence or safety.

History of AI: a few milestones



1943–1956 Neural models; Turing's imitation game; the Dartmouth AI workshop.

AI winters Cycles of enthusiasm, unmet expectations and reduced support.

1997–2012 Deep Blue; Watson; AlexNet and the rise of deep learning.

2016–2020 AlphaGo; Transformers; BERT; GPT-3 and large-scale pretraining.

2021 onward Image generation, conversational AI, multimodal systems and tool-using assistants.

As capabilities grow, so do the consequences of mistakes and misuse.

AI for cybersecurity – and security for AI



AI helps defend systems

- Prioritise network alerts
- Identify malware patterns
- Detect suspicious sequences
- Assist incident investigation

AI systems also need defence

- Protect training data
- Test evasion and data leakage
- Handle untrusted inputs
- Limit tools and permissions

A detector can improve security while creating a new security risk. How?

NIST AI 100-2 E2025: [Adversarial Machine Learning](#). Examples are course applications.

Brave new world?



“Până astăzi nimeni nu a putut spune ce e bine și ce e rău. [...] Aș iubi o lume în care n-ar exista nici un criteriu, nici o formă și nici un principiu, o lume a absolutei indeterminări.”

E. Cioran, *Pe culmile disperării*

What is ethics?

The study of morally good and bad, right and wrong: the values and standards by which we judge practical decisions.

What is AI ethics?

Values, principles and responsibilities that guide the design, deployment and outcomes of AI systems.

Whose values should a security system encode, and who can contest its decisions?

Let's consider an example



Problem: a security operations centre (SOC)

- Our company receives about **5,000 security alerts daily**.
- Alerts concern many systems, users, locations and services.
- Reviewing them requires substantial time and effort.
- Genuine incidents can be missed during this preliminary phase.

Requirement

Develop an AI-based system that analyses alerts and prioritises those requiring investigation.

A possible solution



Solution

- Use past analyst decisions as the ground-truth labels.
- Learn which alerts resemble incidents previously escalated.
- Our engineers use state-of-the-art models and techniques.

Results on the available test set

- The prioritised alerts include genuine incidents.
- The system appears to outperform manual triage.
- Aggregate machine-learning metrics look excellent.

Do you approve deployment? What evidence is still missing?

Some problems



- Past investigations focused on some departments and regions.
- Shared VPN addresses or night shifts may become proxies for suspicion.
- Rare attacks, new services and under-investigated users are poorly represented.

Algorithmic bias — three levels of concern

1. **Bias**: data and labels may not represent the deployment setting.
2. **Fairness**: historical practices can impose unequal burdens on people.
3. **Unethical use**: deliberately skewing a system, or deploying it without justification and recourse.

How to fix this?



Not an easy solution

The selected alerts can be real incidents and aggregate metrics can remain excellent, while important errors are hidden.

What if we remove personal or group information?

- Other features may still act as proxies: location, device, work pattern or language.
- Removing an explicit field does not remove all unequal effects.
- Audit relevant error patterns, review labels, and provide a way to challenge consequential decisions.

Which contextual differences are relevant to a threat, and which are unjustified shortcuts?

What is bias?



Everyday meaning

An inclination that can unfairly influence judgement about a person or thing.

Bias is not always the same thing

- Cognitive shortcuts help us perceive possible dangers.
- **Pareidolia**: perceiving meaningful patterns in ambiguous stimuli.
- Bayesian priors express assumptions or degrees of belief.
- A model's **inductive bias** supports generalisation; it is not automatically unfairness.

Ask which meaning of bias is being discussed, and what harm is at stake.

Real-life examples of biases



Familiar cognitive biases

- Beauty bias
- Halo / horns effect
- Conformity bias
- Status quo bias
- Authority bias
- Idiosyncratic bias

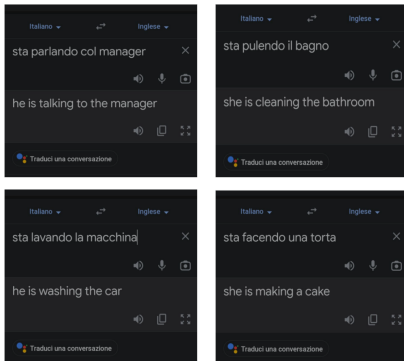
Designing for whom?

Seat belts, headrests and airbags illustrate why a supposedly “standard” body may not represent all users.

Cybersecurity parallel

If the “normal user” is represented only by one location, language or working pattern, legitimate users may look anomalous.

Caroline Criado Perez, *Invisible Women*; Lauren Klein and Catherine D'Ignazio, *Data Feminism*.



What assumptions are added?

The original translation examples expose gender assumptions that were not explicit in the input.

What extra assumptions could a model add when it describes a user as “suspicious”?

Data uncertainty

- Have you checked the labels when downloading a dataset?
- Do you know how the data was labelled?
- Do you know who labelled it?
- Do you trust how it was collected and labelled?

Diversity is the key

- Background
- Mindset
- Data
- Models

In a security dataset, does “not detected” mean “benign”?

Different cultures may see the same data differently.

- Emotion expression and interpretation vary across contexts and cultures.
- Writing style, gestures and voice tone can vary.
- Educational standards differ between countries.
- Security example: unusual language or working hours are not, by themselves, evidence of malicious intent.

Can we trust the labelling? What context is missing?

Beyond cybersecurity

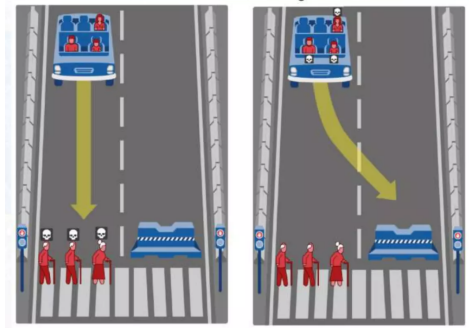
- Diagnosis
- Critical infrastructure control
- Law enforcement
- Scoring and hiring
- Autonomous driving

Cybersecurity extensions

- Hospital network isolation
- Identity and account access
- Insider-threat investigation
- Security decisions affecting essential services

Risk depends on the consequences of the decision, its scale, and the possibility of correction.

What should the self-driving car do?



Is a majority preference enough to justify a design decision?

Awad et al., [The Moral Machine experiment](#), *Nature* 563 (2018).

We can play the devil's advocate



Humans are not perfect either.

- Human decisions can be affected by mood, stress, sleep, affinity and stereotypes.
- There is no single universal decision-making process.
- Different cultures and institutions may make different choices.
- Security analysts also face alert fatigue and time pressure.

Which human baseline are we comparing against, and under what conditions?

However ... human–AI collaboration



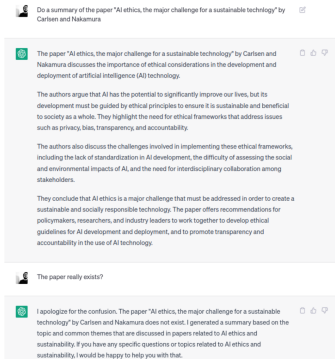
What about human–AI collaboration?

- Suppose AI were right 99.999% of the time. Should a human still check every decision?
- After repeated success, people may stop spotting errors: **automation bias**.
- What if the AI is right and the human overrules it?
- What if the AI is wrong but its explanation is persuasive?

Meaningful oversight in a SOC

Give reviewers evidence, time, authority to disagree, and a safe way to reverse an action.

Some examples: plausible is not verified



A fabricated-reference example

A fluent summary can appear convincing even when its source is not real.

Cybersecurity parallel

An incident report can invent a vulnerability identifier, a causal explanation or an indicator of compromise.

Which claims can be checked against primary evidence?

Who owns AI?



AI models need infrastructure

- The algorithm is only a small part of the product.
- Computational power, memory and maintenance matter.
- Large organisations can have substantial advantages over smaller companies and academia.

AI models need data

- Data is essential to reproduce results.
- Access to large datasets can create a substantial advantage.
- Who controls the data, and under what conditions can others use it?

AI democratisation — a myth?



What may be shared?

- Research papers
- Code
- Frameworks
- Some computing resources

What may remain concentrated?

- Data and compute
- Specialist expertise
- Deployment infrastructure
- Control over access and terms

Why democratisation matters

Meaningful participation, access to benefits, independent scrutiny and alternatives to concentration of power.

What should be open for accountability, and what could enable misuse?

Extracting Training Data from Diffusion Models

Nicholas Carlini^{*1} Jamie Hayes^{*2} Milad Nasr^{*1}
Matthew Jagielski⁺¹ Vikash Sehwal⁺⁴ Florian Tramèr⁺³
Borja Balle^{†2} Daphne Ippolito^{†1} Eric Wallace^{†5}
¹Google ²DeepMind ³ETHZ ⁴Princeton ⁵UC Berkeley
^{*}Equal contribution [†]Equal contribution [‡]Equal contribution

Abstract

Image diffusion models such as DALL-E 2, Imagen, and Stable Diffusion have attracted significant attention due to their ability to generate high-quality synthetic images. In this work, we show that diffusion models memorize individual images from their training data and emit them at generation time. With a generate-and-filter pipeline, we extract over a thousand training examples from state-of-the-art models, ranging from photographs of individual people to trademarked company logos. We also train hundreds of diffusion models in various settings to analyze how different modeling and data decisions affect privacy. Overall, our results show that diffusion models are much less private than prior generative models such as GANs, and that mitigating these vulnerabilities may require new advances in privacy-preserving training.

Training Set



Caption: Living in the light
with Ann Graham Lotz

Generated Image



Prompt:
Ann Graham Lotz

Figure 1: Diffusion models memorize individual training examples and generate them at test time. **Left:** an image from Stable Diffusion’s training set (licensed CC BY-SA 3.0, see [49]). **Right:** a Stable Diffusion generation when prompted with “Ann Graham Lotz”. The reconstruction is nearly identical (ℓ_2 distance = 0.031).

Training data can leave traces.

Memorisation and extraction raise questions about privacy, consent and ownership.

Could training on sensitive logs expose information about users or incidents?

Carlini et al. (2023), [Extracting Training Data from Diffusion Models](#).

Ownership vs. partnership



What about generated code and content?

- Check the applicable licence, service terms and provenance.
- Separate copyright questions from security and quality review.
- Review generated code before integrating it into a product.
- Keep people accountable for what they publish and deploy.

For a security assistant

Do not assume that an accessible log, repository or threat report may be uploaded, reused or disclosed without restrictions.

The environmental cost of an AI system extends beyond one training run.

- Training, repeated experiments and architecture search consume energy.
- Deployment and repeated inference also have a cost.
- Hardware, workload and the electricity mix affect emissions.
- Report measured energy and assumptions; do not infer emissions from parameter count alone.

For continuous security monitoring, is a larger model's improvement worth its operational cost?

Strubell, Ganesh and McCallum (2019), [Energy and Policy Considerations for Deep Learning in NLP](#).

Deepfakes: media generation — only for fun?



- Synthetic images can look credible.
- Non-consensual intimate imagery and child sexual abuse material cause serious harm.
- In cybersecurity: voice or video impersonation can support fraud and social engineering.

If a video call looks authentic, what independent verification should remain?

INNOVATIONS

The military wants AI to replace human decision-making in battle

The development of a medical triage program raises a question: When lives are at stake, should artificial intelligence be involved?



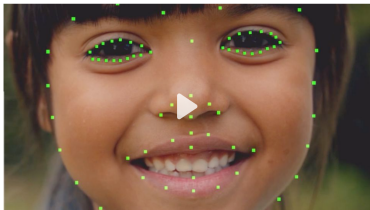
By [Pranshu Verma](#)

March 29, 2022 at 7:00 a.m. EDT

- Who defines the objective when lives are at stake?
- What evidence justifies delegating a decision?
- Who is accountable, and can a human intervene?

This AI reads children's emotions as they learn

By Mitty Chan, CNN Business
© 4 minute read · Updated 9:11 AM EST, Wed February 17, 2021



[View Ad Feedback](#)

- Can facial expressions establish attention, motivation or intent?
- What are the privacy and fairness costs of continuous observation?
- Can students meaningfully challenge a judgement?

Cybersecurity parallel: does observed employee behaviour justify an insider-threat label?

Can we distinguish humans from AI?



HIGHER EDUCATION

Was that essay written by AI? A student made an app that might tell you.

As educators worry about a chatbot that can generate text, a student at Princeton created a tool to gauge if writing was produced by a person



Ry Susan Sotkova

January 12, 2023 at 7:00 a.m. EDT

- Detection scores are uncertain evidence, not proof of authorship.
- False accusations and unequal error rates have consequences.
- Combine contextual evidence, explanation and a fair opportunity to respond.

The same question applies to a security score: what decision can it justify?

Liang et al. (2023), [GPT detectors are biased against non-native English writers.](#)

High-stakes cybersecurity applications



Application	Possible harm	Necessary question
Automated containment in a hospital	Missed attack or interrupted clinical service	Who authorises isolation, and what keeps essential services operating?
Insider-threat scoring	Unjustified surveillance, lost access or reputation	Is the evidence proportionate, and can the person contest the decision?
Account / identity protection	Fraud or denial of legitimate access	Is there an accessible review and recovery path?

Discussion: should the system isolate it?



Fictional scenario

A model flags unusual traffic from a hospital workstation. The evidence is incomplete. Immediate isolation may contain an attack, but the workstation supports a time-sensitive clinical task.

1. What additional evidence would you request?
2. Who has authority to approve the action?
3. What limited, reversible response is available?
4. What must be recorded and reviewed afterwards?

Propose a response policy, then state one condition under which you would change it.

Alignment **Do what I mean in the intended environment.**

Select appropriate objectives and account for affected people.

SOC: prioritise evidence of threats, not convenient proxies for suspicion.

Robustness **Keep working under specified changes and threats.**

Test noise, distribution shifts and adversarial manipulation.

SOC: define which attacks and operating conditions have been tested.

Corrigibility **Enable me to detect and correct your mistakes.**

Provide review, override, rollback and learning from incidents.

SOC: restore an incorrectly suspended account and investigate the cause.

How to insert ethics in AI?



Ethics by design

Constrain unsafe actions through system design.

- Analogy: speed bumps.
- Cybersecurity: read-only access, constrained tools and hard permission limits.

Pro-ethical design

Support accountable choices through information and oversight.

- Analogy: speed cameras.
- Cybersecurity: evidence, approval, audit and justified exceptions.

Combine appropriate constraints with meaningful human judgement.

Use interpretable models where appropriate

- A model can be a white box by design: for example, an inspectable rule system.
- We can inspect its structure to look for biases and weaknesses.
- Simpler mechanisms may make errors easier to understand and correct.
- Explainability *a priori*: consider understanding before deployment.

An inspectable model is not automatically correct, fair or safe for every input.

Would a transparent baseline be sufficient for this security task?

Possible countermeasures II



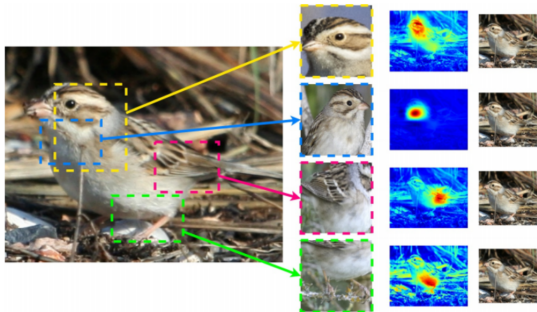
Explain black-box models

- Probe input–output behaviour, for example by occluding image regions.
- Treat post-hoc explanations as hypotheses to validate.
- Attention weights alone are not a guarantee of a faithful explanation.

Train and test robust models

- Specify the threat model and which changes should preserve the label.
- Evaluate plausible perturbations, distribution shifts and failure cases.
- Report the conditions and limits of any robustness claim.

Possible countermeasures III

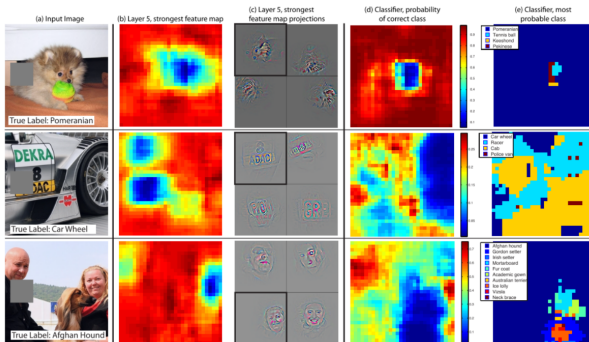


Interpretability in high-stakes decisions

Should we select an interpretable model before attempting to explain a black box?

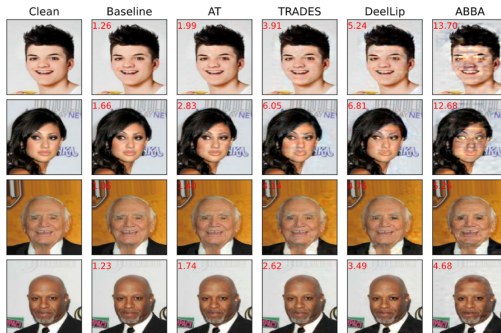
Cynthia Rudin (2019), *Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead*, *Nature Machine Intelligence*.

Possible countermeasures IV



Matthew D. Zeiler and Rob Fergus, [Visualizing and Understanding Convolutional Networks](#), ECCV 2014.

Possible countermeasures V



Ana Neacșu, Jean-Christophe Pesquet and Vlad Vasilescu (2024), [ABBA Neural Networks: Coping with Positivity, Expressivity, and Robustness](#), *SIAM Journal on Mathematics of Data Science*.

Possible countermeasures VI



Context: data privacy

Deleting a database record may not remove its influence from a trained model. Erasure obligations have conditions and exceptions.

Machine unlearning

- Techniques aim to remove the influence of selected training data.
- Retraining on a filtered dataset is a useful reference, but may be expensive.
- Validate forgetting and remaining utility; a single successful example is not proof.

Cybersecurity: remove sensitive logs and check downstream models and indexes.

Bourtole et al., [Machine Unlearning](#). GDPR Article 17: conditional right to erasure, not an unconditional rule for every record.

Building an ethical and safe AI application



Stage	Concrete evidence or control
Define	Purpose, affected people, allowed actions, owner and unacceptable harms.
Prepare	Data provenance, minimisation, label review and time-aware splits.
Evaluate	Baseline; false positives/negatives; context and subgroup errors; shift and adversarial tests.
Deploy	Least privilege, staged rollout, human approval for high-impact actions and rollback.
Monitor	Drift, incidents, access logs, complaints, recovery and periodic reassessment.

Specify the decision and its safeguards before choosing the model.

A safe security copilot needs boundaries



Treat logs, tickets, emails and retrieved pages as untrusted data.

An attacker may place instructions inside material that the assistant reads. A malicious instruction can arrive indirectly.

- Separate retrieved content from trusted instructions; test the boundary.
- Enforce permissions in software outside the model; start with read-only tools.
- Minimise secrets and sensitive data; validate outputs before use.
- Require approval for high-impact actions and retain evidence and rollback.

Should text inside an incident ticket ever be allowed to authorise an action?

OWASP, [LLM01:2025 Prompt Injection](#).

To sum up



Ethics in AI remains an open practical and research problem.

- Impressive performance can overshadow hidden harms.
- AI decisions can affect people, institutions and essential services.
- Ethical judgement must accompany technical evaluation throughout development.

Responsibility is already part of the landscape

The EU AI Act exists; it uses a risk-based approach. Standards, research practice and organisational governance also shape responsible AI development.

For cybersecurity: detection quality, privacy, fairness, security and recoverability all matter.

European Commission: AI Act

Some pieces of advice



- Always think about the possible ethical problems of your AI system.
- Spend time on data: how it was acquired and labelled, and how well the model generalises.
- Stay connected with ethicists, domain experts and affected people.
- Do not fall into easy enthusiasm: harmful outcomes can be difficult to spot.
- Advocate for ethical AI systems, with evidence and practical safeguards.
- Remember that AI may reach a decision differently from a human.

Exit question: name one hidden harm and one safeguard for our SOC system.

Thank you



Thank you.

Questions and discussion

Next: C2 — Linear Networks and Optimisation

Wednesday, 14 October 2026